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Multimodal Physics-Based Aging Model for Life Prediction of Li-Ion Batteries

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An isothermal, multimodal, physics-based aging model for life prediction of Li-ion batteries is developed, for which a solvent-decomposition reaction leading to the growth of a solid electrolyte interphase (SEI) at the carbonaceous anode material is considered as the source of capacity fade. The rate of SEI film growth depends on both solvent diffusion through the SEI film and solvent-reduction kinetics at the carbon surface. The model is able to simulate a wide variety of battery aging profiles, e.g., open-circuit and constant-voltage storage, charge/discharge cycling, etc. An analysis of capacity-fade data from the literature reveals that the same set of aging parameters may be used for predicting cycling and constant-voltage storage. The use of this set of parameters for predicting storage under open-circuit voltage points out that part of the self-discharge is reversible.
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Aging of Li-ion cells is becoming more and more a concern because of their potential use in long-life applications such as electric vehicles (EVs) and hybrid EVs, for which lifetimes of 10 and 15 years are expected, respectively.[1] In terms of cycle life, a lifetime up to 3000 deep cycles can be requested for high-energy applications (typical for EVs). As a result of these drastic conditions, there is a need for lifetime-prediction methods that cannot be substituted for test programs. Among those methods, some are based on a purely statistical approach, like the standard Wohler method,[2] while others are based on a physical description of the aging phenomena and are referred to as physics-based models.

Physics-based models fall into two groups. Some of them are referred to as "phenomenological," because aging sources [e.g., a side reaction leading to the growth of a solid electrolyte interphase (SEI)], explicitly show up in the set of governing equations and are in direct interaction with the other model features. The second group gathers mathematical models that are essentially similar to those for a pristine cell, and some of the model parameters (e.g., SEI film resistance or thickness, volume fraction of active material, etc.) are updated using some isolated empirical relations or curve-fitting procedures.

Phenomenological physics-based models constitute the most sophisticated but also the most complex approach, because an extensive understanding of the aging mechanisms is required. Once the model is validated, it allows for an analysis of a wide variety of operating conditions and control strategies and can even be used by battery manufacturers for improving the battery design. So far, most of the physics-based models have embedded only a single aging source due to both the extreme complexity of real systems and the lack of theoretical background in the modeling of some aging sources like the structural degradation of active materials.

Let us give a brief overview of the phenomenological models reported so far in the literature for battery aging. Darling and Newman made a first attempt to model side reactions in lithium-ion batteries by incorporating a solvent oxidation reaction into a lithium-ion battery model.[3] Even though a simplified treatment of the oxidation reaction was used, their model was able to draw several interesting conclusions about self-discharge processes in the cells under consideration and the impact on positive electrode state-of-charge. Ramadass et al. developed a capacity-fade model by ascribing the loss of active lithium during charge/discharge cycling to a continuous SEI film formation over the surface of the negative electrode.[4] No transport limitation for the solvent across the SEI was considered, i.e., the solvent was assumed to be abundant at the anode surface. Ploehn et al. presented a "solvent-diffusion" model for capacity-fade predictions during storage under constant voltage at various temperatures.[5] They came out with a correlation relating the fractional capacity loss to the square root of storage time. In this study, the kinetics for the side reaction was not considered. Ramasamy et al. developed a mathematical model that predicts the capacity loss of the carbon electrode during open-circuit voltage (OCV) storage due to a continuous reduction of organic solvent and concomitant de-intercalation of lithium from the carbon.[6] Their model was able to predict the anode potential rise during OCV storage. Just like in the paper by Ramadass,[4] no transport limitation across the SEI was taken into account.

In almost all the above-mentioned studies but the work by Darling and Newman,[3] the solvent decomposition at the anode surface was considered as the side reaction that dominates the capacity fade of the cell. Furthermore, all of the studies reported so far were limited to the description of the capacity fade of the battery in a single mode of operation, and the choice of neglecting either the solvent transport across the SEI or the side reaction kinetics was not obvious.

Having a physics-based model with the ability to simulate the cell performance and aging under different modes of operation (i.e., multimodal) is much desired for the purpose of battery-life prediction. Furthermore, experimental data of different types (charge/discharge cycling, OCV storage, constant voltage storage, etc.) can be used to refine the set of input parameters for the model, thus greatly improving its reliability, predictability, and applicability.

In this study, a physics-based phenomenological multimodal model is developed for battery-life prediction, with the aging consisting in a solvent-decomposition side reaction leading to the growth of an SEI film at the carbonaceous anode material. The model is used to analyze capacity-fade data of $LiCoO_2$/graphite Li-ion cells obtained under several aging procedures (charge/discharge cycling, constant-voltage, and OCV storage). Finally, the model is used to assess the influence of the SEI porosity on capacity fade, and the SEI growth law is studied under two storage modes, namely, constant-potential and OCV.

Model Development

Physical picture of SEI growth.— In the case of graphite anodes in Li-ion batteries, there are continuous small-scale surface reactions between lithiated graphite and solution species that lead to a gradual decrease in cyclable lithium and to an increase in the electrode impedance upon cycling.[7-9] These small-scale reactions take place when the SEI film has an "open structure" that can be due to a nonoptimized SEI formation step during the first few cycles, unavoidable volume changes in the graphite particles during lithium insertion–deinsertion, SEI-film partial dissolution, side reactions leading to gas evolution, etc.

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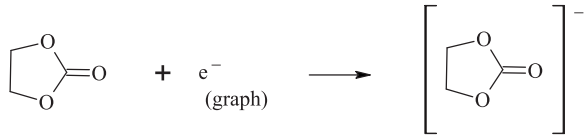

Scheme 1.

Regarding the solvent-reduction mechanisms [for example, for ethylene carbonate (EC) or propylene carbonate] Dey initially proposed a two-electron reduction process, bringing about a precipitate of Li_2CO_3 in the organic electrolyte and ethylene or propylene gas.[10] After extensive studies of solvent- and salt-reduction processes in the 1990s, Aurbach et al. concluded that lithium ethylene dicarbonate $(CH_2OCO_2Li)_2$ and ethylene gas resulting from a one-electron reduction process of EC are the dominant products; however, the dicarbonate is highly sensitive to water traces and rapidly reacts with them to form Li_2CO_3.[11] Yang et al. also investigated the passive SEI-film composition on the carbon anode surface in contact with an EC-based electrolyte using Fourier transform infrared and mass spectroscopy.[12] They observed that the passive SEI film contains chiefly $(CH_2OCO_2Li)_2$ not only for a single EC solvent but also for binary EC/diethylcarbonate (DEC) (1:1) regardless of the EC concentration. Only EC is decomposed in a binary solvent consisting of EC/DEC or EC/dimethylcarbonate (DMC). DEC and DMC do not seem to be decomposed in the presence of EC. Using high-level density functional calculations, Wang et al. concluded that reductive decomposition of EC initially proceeds through an ion-pair intermediate.[13] It then homolytically cleaves via an 11.5 kcal/mol barrier to generate a radical anion, which undergoes a secondary reaction by barrier-free self-dimerizing.

In this study, the SEI is considered to be made of lithium ethylene dicarbonate $(CH_2OCO_2Li)_2$. The rate-determining step (RDS) is considered to be the radical anion formation as seen in Scheme 1.

Then, as in Scheme 2 and 3, several possible follow-up reactions may occur that lead to the formation of the ethylene dicarbonate anion. Finally, the precipitation of solid lithium ethylene dicarbonate occurs as seen in Scheme 4.

The SEI film on the negative electrode is generally recognized as having a complex structure consisting of a thin dense part in the vicinity of the particle surface of a few nanometers and a thicker secondary porous part exposed to the solution. It is considered that the outer part of the SEI, for which the pores are filled with electrolyte, controls the progress of SEI growth and, consequently, the extent of capacity fade and impedance rise. The porous structure of the SEI film allows for the permeation of solvent molecules toward the active particle surface and therefore controls the rate of solvent decomposition reaction. Some previous works consider the transport of electrons through the SEI as the rate-determining process for SEI growth.[14,15] Using this assumption, Peled derived a parabolic-growth law for the SEI on lithium-metal electrodes.[14] The partial electronic conductivity of the SEI was ascribed to the deposition of lithium metal within the bulk of the SEI due to local excessive heating and the high electric field within the film.[14] In the case of SEIs on graphite, lithium-metal deposition within the SEI is less prone to occur, and therefore, as a first approximation, no electron transport through the SEI is considered, except perhaps in the thin inner part of the SEI in contact with the graphite for which electron tunneling is possible.

From the above considerations, let us describe the physical picture of the SEI we defined. We decided to model the SEI as a single phase covering uniformly the surface of the carbon particles. Such an assumption has been extensively used for ion-exchange membrane modeling, for instance, even though a more accurate treatment would require the consideration of a two-phase system.[16,17] The main advantage is that it considerably reduces the number of governing equations and input parameters, which is of interest because of the lack of input parameter values and because we want to keep the model simple enough for fast convergence. It is composed of $(CH_2OCO_2Li)_2$ as the major component, plus minor species: EC, ethylene, and defects that act as charge carriers (lithium vacancies V_{Li}^- and interstitial Li_i^+), etc. Dilute solution theory is used to describe the transport properties of the minor species. For the sake of simplicity, the entire SEI is assumed to be electronically insulating. A schematic of the physical picture for the model is shown in Fig. 1.

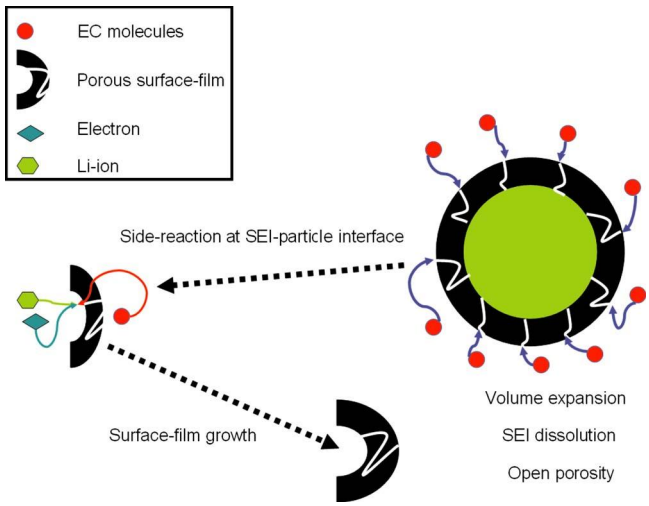

Figure 1. (Color online) Physical picture for SEI growth at the anode particle surface.

Governing equations.— For aging simulation purpose, the use of simplified models saves a lot of time in solving procedures, because time-consuming simulations for thousands of cycles or several years of storage need to be run. Therefore, and also for the sake of simplicity, a single particle (SP) model was used in this study. The SP approach was originally developed by Atlung et al.,[18] used by Haran et al.[19] for the metal-hydride battery, and extended to the lithium-ion battery later on. A detailed description of this model can be found in the work by Santhanagopalan et al.[20] SP models do not consider the transport limitations in the liquid phase and therefore are not accurate for high charge/discharge rates.

Li insertion/deinsertion in the active materials occurs through a surface reaction that takes place at the interface between the graphite and the SEI and that follows Butler–Volmer kinetics

$$i_{int} = Fk_{int}^{0,app}c_{Li}^{\beta}(c_{Li}^{max} - c_{Li})^{1-\beta}\left\{\exp\left(\frac{(1-\beta)F}{RT}(\Phi_1 - U - R_{SEI}i_t)\right) - \exp\left(-\frac{\beta F}{RT}(\Phi_1 - U - R_{SEI}i_t)\right)\right\} \quad [1]$$

The variables are as defined in the list of symbols. The potential of the ion-conductive phase Φ_2 at the interface between the graphite and the SEI (where the reaction takes place) is related to the potential of the electrolyte phase (taken to zero because of the SP assumption) using Ohm's law and is thus equal to $R_{SEI}i_t$, with $R_{SEI}(t) = \delta(t)/\kappa_{SEI}$ the ionic resistance of the SEI. Note that because we assume a constant ionic conductivity κ_{SEI} throughout the SEI, the resistance increase is solely dependent on the thickness increase. The expressions for the equilibrium potentials of cathode and anode are given in the Appendix.

Solid-state lithium diffusion in the spherical active particles follows Fick's second law

$$\frac{\partial c_{Li}}{\partial t} = \frac{1}{r^2}\frac{\partial}{\partial r}\left(D_{Li}r^2\frac{\partial c_{Li}}{\partial r}\right) \quad [2]$$

Two boundary conditions are required for this equation. One is deduced from symmetry at the center of the spherical active particles

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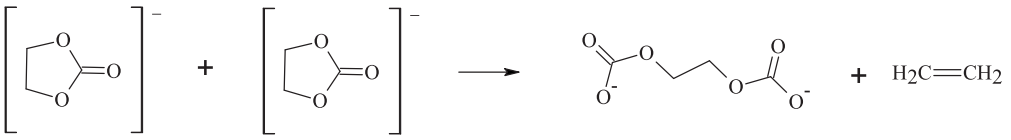


Scheme 2.

$$\left.\frac{\partial c_{\mathrm{Li}}}{\partial r}\right|_{r=0} = 0 \qquad [3]$$

and the other results from relating the lithium flux density at the particle surface to the intercalation current density

$$\left.\frac{\partial c_{\mathrm{Li}}}{\partial r}\right|_{r=R} = -\frac{i_{\mathrm{int}}}{FD_{\mathrm{Li}}} \qquad [4]$$

The RDS for the formation of the SEI product was assumed to be the formation of EC radical anions through Scheme 1. This reaction takes place at the same interface as Li insertion/deinsertion reaction, and the kinetic expression for this side reaction reads

$$i_{\mathrm{s}} = -Fk_{\mathrm{f,s}}c_{\mathrm{EC}} \exp\left[-\frac{\beta_{\mathrm{s}}F}{RT}(\Phi_1 - R_{\mathrm{SEI}}i_{\mathrm{t}})\right] \qquad [5]$$

As the radical anion is consumed quickly to form irreversibly the final SEI product through the possible reaction schemes described above, it seems reasonable to use a Tafel-like expression. Note, however, that Eq. 5 does not have a regular form, as the exponential term does not show the presence of the equilibrium potential. This alternative form was chosen to have only a single adjustable parameter $k_{\mathrm{f,s}}$ instead of two (an exchange current density and an equilibrium potential). Equilibrium potentials are not well-known for those side reactions, even though some potential ranges were reported in the literature for the SEI formation reaction. Mori et al. reported the main SEI formation step to be occurring between 0.55 and 0.2 V vs Li electrode.[21] Aurbach et al. stated that the major reduction processes occur at around 1 V vs Li electrode and continue gradually as the potential is lowered.[9] Naji et al. reported that the formation of an EC radical anion occurs below 0.8 V vs Li electrode.[22] In almost all the previous modeling works, a value of 0.4 V vs Li electrode was chosen as the equilibrium potential of the SEI-formation side reaction at the anode.[4,23-26]

The current balance for the negative electrode also should be updated to include the side-reaction current density (i_{s} in Eq. 5)

$$i_{\mathrm{t}} = i_{\mathrm{int}} + i_{\mathrm{s}} \qquad [6]$$

i_{t} is the total current density in the negative electrode (which refers to the total interfacial surface area of the anode) and i_{int} stands for the part of current density that takes part in the intercalation process.

Next, the transport of solvent molecules (EC) through the SEI film is addressed. The SEI is considered to have a planar geometry, as its dimension is significantly smaller than that of active-material particles. The material balance of solvent (EC) inside the SEI film takes the following form

$$\frac{\partial c_{\mathrm{EC}}}{\partial t} = -\frac{\partial N_{\mathrm{EC}}}{\partial r} \qquad [7]$$

with the solvent flux density defined as

$$N_{\mathrm{EC}} = -D_{\mathrm{EC}}\frac{\partial c_{\mathrm{EC}}}{\partial r} + \nu c_{\mathrm{EC}} \qquad [8]$$

The second term on the right side of Eq. 8 represents the convection term, with ν the SEI film velocity defined as

$$\nu = \frac{d\delta}{dt} \qquad [9]$$

with δ the SEI film thickness. Substitution of Eq. 9 into Eq. 8 and of Eq. 8 in Eq. 7 leads to the final form of the material balance on solvent

$$\frac{\partial c_{\mathrm{EC}}}{\partial t} = D_{\mathrm{EC}}\frac{\partial^2 c_{\mathrm{EC}}}{\partial r^2} - \nu\frac{\partial c_{\mathrm{EC}}}{\partial r} \qquad [10]$$

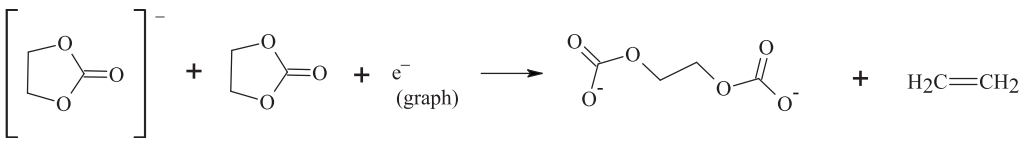
The boundary condition on the solvent flux at the particle–SEI interface ($r = R$) in the case where the follow-up reaction is the coupling between two radical anions (Scheme 2) reads

$$N_{\mathrm{EC}}|_{r=R} = \frac{i_{\mathrm{s}}}{F} \qquad [11]$$

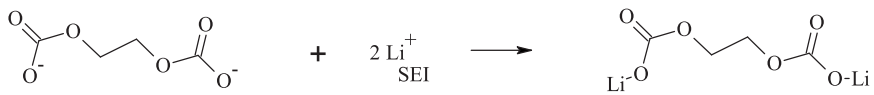
The rate of growth of the SEI layer due to the accumulation of lithium ethylene dicarbonate is related to side-reaction current density through

$$\frac{d\delta}{dt} = -\frac{i_{\mathrm{s}}}{2F}\frac{M_{\mathrm{SEI}}}{\rho_{\mathrm{SEI}}} \qquad [12]$$

Another boundary condition needs to be set at the SEI film–electrolyte interface ($r = R + \delta$). Assuming the abundance of electrolyte at this interface and no change in electrolyte composition during the battery life, it is reasonable to have a constant solvent concentration at this boundary. This concentration is not known and depends on the physical properties of the SEI film. Because we have chosen to describe the SEI as a single phase, the solvent concentration (EC) in the SEI c_{EC} at the SEI/electrolyte interface is related to



Scheme 3.



Scheme 4.

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that in the electrolyte $c_{EC,sln}$ through a partition coefficient which is a thermodynamic quantity

$$K_{EC} = \frac{c_{EC}|_{r=R+\delta}}{c_{EC,sln}} \quad [13]$$

However, the value of such a partition coefficient is not known. Although the SEI film is taken as a single phase, in order to get a feel for the range of c_{EC}, we can use the following trick. Let us consider the SEI as a two-phase medium, where the liquid electrolyte fills the porosity ε_{SEI} of the SEI. In this case, the volume-averaged EC concentration in the SEI would be $\varepsilon_{SEI}c_{EC,sln}$. Comparing the two physical pictures, the partition coefficient in the SP model can be regarded as the SEI porosity in the two-phase model

$$K_{EC} = \varepsilon_{SEI} \quad [14]$$

As initial conditions, a flat lithium concentration profile corresponding to the initial material utilization was set in the active particles and a flat solvent concentration equal to $K_{EC}c_{EC,sln}$ was set in the SEI.

Differential algebraic equations were converted into dimensionless forms and cast in finite-difference forms. The Crank–Nicholson implicit method was used to evaluate the time derivatives. All the equations were solved simultaneously using the FSOLVE subroutine embedded in the MATLAB software. A summary of the governing equations and boundary conditions is presented in Table I.

Table I. Summary of the equations used in the aging model.

Governing equations

$$i_{int} = Fk_{int}^{0,app}c_{Li}^{\beta}(c_{Li}^{max} - c_{Li})^{1-\beta}\left\{\exp\left[\frac{(1-\beta)F}{RT}\left(\Phi_1 - U - \frac{\delta}{\kappa_{SEI}}i_t\right)\right] - \exp\left[-\frac{\beta F}{RT}\left(\Phi_1 - U - \frac{\delta}{\kappa_{SEI}}i_t\right)\right]\right\}$$

$$\frac{\partial c_{Li}}{\partial t} = \frac{1}{r^2}\frac{\partial}{\partial r}\left(D_{Li}r^2\frac{\partial c_{Li}}{\partial r}\right)$$

$$i_s = -Fk_{f,s}c_{EC}\exp\left[-\frac{\beta_s F}{RT}\left(\Phi_1 - \frac{\delta}{\kappa_{SEI}}i_t\right)\right]$$

$$i_t = i_{int} + i_s$$

$$\frac{\partial c_{EC}}{\partial t} = D_{EC}\frac{\partial^2 c_{EC}}{\partial r^2} - \frac{d\delta}{dt}\frac{\partial c_{EC}}{\partial r}$$

$$\frac{d\delta}{dt} = -\frac{i_s}{2F}\frac{M_{SEI}}{\rho_{SEI}}$$

Boundary conditions

$$\frac{\partial c_{Li}}{\partial r}\Big|_{r=0} = 0$$

$$\frac{\partial c_{Li}}{\partial r}\Big|_{r=R} = -\frac{i_{int}}{FD_{Li}}$$

$$-D_{EC}\frac{\partial c_{EC}}{\partial r}\Big|_{r=R} + \nu\, c_{EC}|_{r=R} = \frac{i_s}{F}$$

$$c_{EC}|_{r=R+\delta} = \varepsilon_{SEI}c_{EC,sln}$$

Results and Discussion

In this section, the model is used to study the capacity fade in different modes of operation, namely, charge/discharge cycling, constant-voltage storage, and OCV storage. The $LiCoO_2$/graphite chemistry was picked up as the case study. Lithium-insertion rate constant and lithium-diffusion coefficient for both electrodes were refined from charge/discharge cycling data (0.5 C) from a pristine 18650 Sony cell[27] and are given in Table II. These values were unchanged in the following aging data analysis. Note that the rate-constant values are generally slightly underestimated compared to a full model[27] because of the SP assumption of no liquid-phase transport limitations. The equilibrium potential functions for $LiCoO_2$ and graphite were taken from Ref. 27 and are reproduced in the Appendix. In the following, only the parameters dealing with the SEI growth on anode, i.e., the side-reaction rate constant $k_{f,s}$ and the solvent-diffusion coefficient D_{EC}, were refined. The charge-transfer coefficient, β_S, was set arbitrarily to 0.5, as for the lithium insertion reaction. The physical properties of the SEI product, $(CH_2OCO_2Li)_2$, are taken from the work by Borodin et al.[28] They used a quantum-chemistry-based many-body polarizable force field to predict the physical properties of lithium ethylene dicarbonate and lithium methyl carbonate. Physical properties for $(CH_2OCO_2Li)_2$ used in the simulations are presented in Table II. The initial SEI resistance was taken to be 0.001 Ω m^2 based on the work by Zhang et al.[29] This value of resistance corresponds to a thickness of 5 nm based on the ionic conductivity of $(CH_2OCO_2Li)_2$, which seems reasonable for an electrode after a few formation cycles. Notice that the presence of a thin inner SEI layer introduced above, which is usually 1.5–2 nm thick according to Christensen and Newman,[15] is discarded in this work. The SEI porosity is another parameter that needs to be set. Based on preliminary adjustments, a value of 5% was chosen. This parameter reflects the quality of the SEI formed and also is subject to variations upon aging of the battery. Those variations are not considered in the simulations and could be the subject of future studies. The bulk concentration of EC molecules in the electrolyte also needs to be set in the model. There were no data for EC concentration for the Sony 18650 cell; however, Ploehn et al. estimated the EC concentration in an EC–DEC–DMC (1:1:1) electrolyte with 1 M $LiPF_6$ to be 4.541 mol/L.[5] This value was used for the EC concentration in the bulk of electrolyte for all the simulations. The model parameters for the SEI discussed above are gathered in Table II.

Capacity fade during charge/discharge cycling.— There are only a little long-term aging data for Li-ion batteries in the literature. The Sony 18650 cell with the $LiCoO_2$/graphite chemistry (rated capacity of 1.8 Ah) has been the most studied, especially for modeling aging. Most of the aging data for this chemistry were obtained under charge/discharge cycling (0.5C with a potential range 2–4.2 V and a conventional constant current–constant voltage charge) rather than under storage or other aging profiles. Before getting into simulations, we try to estimate the anode contribution to the total capacity fade of Sony 18650 cells. Ramadass et al. quantified the various contributions to capacity fade of Sony 18650 batteries for various cycle numbers and temperatures but not for each electrode.[30] The total capacity loss of 539 mAh after 800 cycles at room temperature is split into 71 mAh due to rate-capability loss, 392 mAh due to secondary capacity loss (structural degradation of active material), and 76 mAh due to primary capacity loss (active lithium consumed in side reactions). See Ref. 30 for the detailed calculation methodology. It is reasonable to assume that the primary capacity loss

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Table II. Model parameters for the two commercial cells and the three aging modes used in this study (default values unless otherwise stated).[a]

	Sony 18650 cell		SAFT MP prototype
	Anode	Cathode	Anode
R (μm)	2 (Ref. 27)	2 (Ref. 27)	1[es]
c_{Li}^{max} (mol/m^3)	30555 (Ref. 27)	51555 (Ref. 27)	30555 (Ref. 27)
A (m^2)	4.38 (Ref. 27)	4.76 (Ref. 27)	21.5 (Ref. 5)
D_{Li} (m^2/s)	2×10^{-14} [ad]	1×10^{-14} [ad]	2×10^{-14} [ad]
$k_{int}^{0,app}$ (m/s)	2.07×10^{-11} [ad]	1.04×10^{-11} [ad]	1.04×10^{-11} [ad]
β	0.5[a]	0.5[a]	0.5[a]
x^0 and y^0	0.74 (cycling) (Ref. 27) 0.99 (OCV storage) (Ref. 6)	0.5 (Ref. 27) (cycling)	0.9[es] (constant-potential storage)
	Anode SEI parameters		
$c_{EC,sln}$ (mol/L)	4.541 (Ref. 5)		
R_{SEI}^0 (Ω m^2)	0.001 (Ref. 29)		
M_{SEI} (kg/mol)	0.162 (Ref. 28)		
ρ_{SEI} (kg/m^3)	1690 (Ref. 28)		
δ^0 (nm)	5[es]		
κ_{SEI} (S/m)	5×10^{-6} (Ref. 28)		
ε_{SEI}	0.05[a]		
β_s	0.5[a]		
D_{EC} (m^2/s)	6.8×10^{-21} [ad] (diffusion-limited case) 3.7×10^{-19} [ad] (adjusted from anode potential rise during OCV storage)		
$k_{f,s}$ (m/s)	1.36×10^{-12} [ad] (kinetic-limited case) 1.36×10^{-7} [ad] (adjusted from anode potential rise during OCV storage)		

[a] ad: adjusted; a: assumed; es: estimated.

(76 mAh) in the cell comes mostly from the side reactions at the anode as shown in various studies,[7,8,30-33] which implies that no lithium-based passivating SEI films are formed at the cathode. Because structural changes can often be neglected for the graphite, we assume that the secondary capacity loss is attributed completely to the cathode. Assuming the fraction of capacity losses due to the decrease in rate capability for each electrode to be proportional to the interfacial resistance increase for the same (i.e., 11- and 16-fold increase for anode and cathode as measured by Zhang et al. by three-electrode impedance spectroscopy in the same aging conditions, respectively[29]), 41% of the total capacity loss due to rate-capability decrease (71 mAh) is attributed to the anode part (29 mAh). In the anode, it is reasonable to think that this source of capability loss is connected to an SEI growth. Because we did not consider any lithium consumption at the cathode, the fraction of capacity loss due to a decrease in rate capability (42 mAh) could be due to a surface decomposition of the layered material, leading to thin insulating layers at the surface of the particles (secondary capacity loss) or to particle disconnection from the electron conductive matrix of the electrode. To summarize, given our assumptions, the anode contribution to capacity losses would be (76 + 29)/539 = 19.5% of the total capacity loss after 800 cycles, which itself corresponds to 31% of the initial capacity. The fraction of capacity loss due to the anode is thus estimated to 0.195 × 0.31 = 6%. A summary of this capacity fade analysis is presented in Table III.

In the published capacity-fade models for charge/discharge cycling so far, the SEI growth in the anode was considered to occur through a side reaction following a Tafel kinetic expression having an equilibrium potential equal to 0.4 V. Transport of solvent molecules across the SEI was simply neglected. In other words, the SEI growth was assumed to be only kinetic-limited.[4,23-26] In the model we developed, however, both kinetic and diffusion limitations are considered.

The objective of the parameter refinement is to account for ca. 6% capacity fade together with an increase of ca. 11-fold in the SEI resistance after 800 cycles at room temperature. This refinement process is not straightforward and is time-consuming; therefore, the parameters reported hereafter are not necessarily the best ones. Simulations were carried out for the two extreme cases of diffusion- and kinetic-limited SEI growth using the parameter values given in Table II. The discharge profiles for various cycle numbers are plotted in Fig. 2, and the cell capacity and SEI resistance as a function of the cycle number is shown in Fig. 3. The SEI resistance increase with time is linear for the kinetic-limited regime and quadratic for the diffusion-limited case. However, both regimes satisfy the optimization objectives, and therefore it is not possible to distinguish whether the SEI growth is more under diffusion or kinetic control at this point of analysis. This may be possible if more experimental data for the anode contribution to the capacity fade are available.

Capacity fade upon storage.— Physics-based models of the aging of lithium-ion batteries during storage were reported by Ramasamy et al.[6] and Ploehn et al.[5] The first authors used a kinetic expression for the solvent reduction at the carbon particle/SEI interface to describe the self-discharge of a graphite anode from a Sony 18650 lithium-ion cell. Solvent molecules and lithium were assumed to be in excess for the solvent-decomposition reaction. Their model was able to account for the anode potential rise during OCV storage of the cell, in agreement with the experimental data. Ploehn et al.[5] developed a "solvent-diffusion" model to describe the SEI growth in SAFT batteries during constant-voltage storage. In contrast with the work by Ramasamy et al.,[6] they considered the SEI growth to be

Table III. Contributions to capacity fade of a Sony 18650 cell after 800 cycles at room temperature. Data for complete cell are taken from Ref. 30.

	Secondary capacity loss (loss of active material) (mAh)	Capacity loss due to rate-capability decrease (mAh)	Primary capacity loss (active lithium loss) (mAh)
Complete cell	392	71	76
Anode	0	29	76
Cathode	392	42	0

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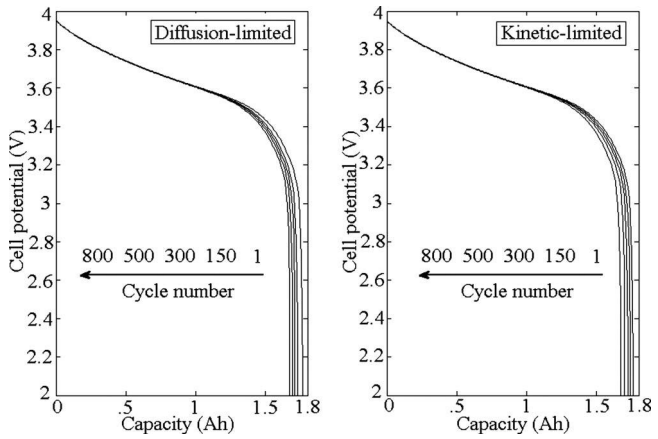

Figure 2. Simulations of the discharge profiles during cycling of a Sony 18650 cell at room temperature and 0.5C rate for kinetic- and diffusion-limited regimes.

limited by the solvent diffusion, and no kinetic expression for the solvent decomposition was considered. Although the two above-mentioned models successfully fitted the experimental data, in both cases some of the assumptions made were not proved to be realistic, such as the choice of a pure kinetic limitation or a pure diffusion limitation of the SEI growth. In order to get more insight in the capacity fade during storage conditions, some of the data reported in those two references were reanalyzed using our model.

Constant-voltage storage.— The parameters refined for aging during cycling and discussed in the previous section are used to simulate aging under storage at constant potential, with the aim of sensing how different the capacity fade between the two different aging modes can be and whether a universal set of side-reaction parameters can describe the capacity fade irrespectively of the operation mode.

The capacity loss of a 5 Ah prismatic "MP" ($LiCoO_2$/graphite) prototype cell was reported by Broussely et al.[32] for constant-voltage storage conditions (3.9 V) at 30°C. The cell specifications for this battery were not reported by Broussely et al.; instead, Ploehn et al.[5] estimated some parameters like the carbon surface area, the anode current collector surface area, etc. Similar to the paper of Ploehn et al.,[5] it is assumed that the capacity fade is entirely due to an SEI growth on the anode. For the sake of simplicity, although the state of charge (SOC) of the cathode is expected to vary because of

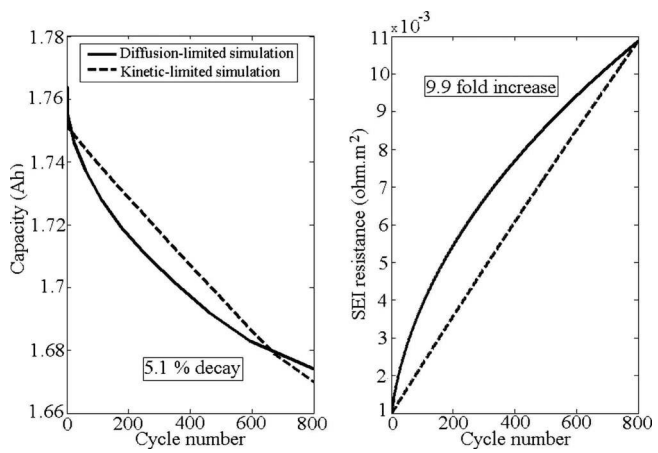

Figure 3. Simulations of capacity fade and SEI resistance increase during cycling of a Sony 18650 cell at room temperature and C/2 for kinetic- and diffusion-limited regimes.

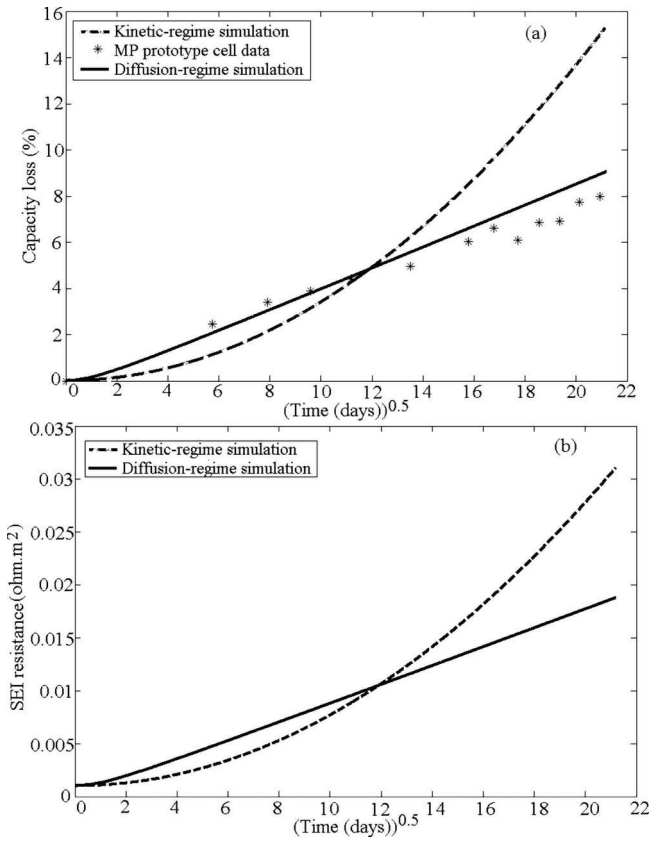

Figure 4. Experimental and simulated capacity fade (a) and simulated SEI resistance (b) of an MP prototype cell under constant-voltage storage (3.9 V) at 30°C for kinetic- and diffusion-limited regimes (data from Ref. 5).

aging when the cell voltage is set, it is assumed here that this does not affect the cathode potential significantly, and therefore simulations were run for the graphite half-cell only (i.e., the graphite electrode potential was set). Using a trial-and-error method as well as the equilibrium potential functions introduced in Appendix A, the constant cell voltage of 3.9 V was estimated to correspond to an initial SOC of 90%, which leads to an open-circuit potential of 0.0664 V vs Li electrode for the graphite electrode. The percentage of capacity loss was calculated by considering the initial anode capacity to be equal to the theoretical one. The estimated and assumed cell parameters used in the simulation of the constant-potential storage of the graphite are summarized in Table II. The values for side-reaction parameters are the same as those refined from the cycling mode. The simulation results for 450 days of storage are overlaid to the experimental data in Fig. 4a. The diffusion-limited case is in much better agreement with the experimental data as compared with the kinetic-limited case. This might be proof that the SEI growth is more under diffusion control. Furthermore, the suitability of the model parameters (diffusion-limited case) from cycling analysis to the constant-potential storage situation tends to prove that aging proceeds through the same mechanism regardless of the mode.

The SEI resistance increase during storage is displayed in Fig. 4b. A 16-fold increase in SEI resistance during the 450 days of storage is observed for the diffusion-limited case. A deeper understanding of the capacity-fade process can be obtained by looking at the current densities for lithium insertion/deinsertion and SEI growth (Fig. 5). The magnitudes are small and, for more clarity, the log of absolute values of current densities are plotted. Note that the current density for the side reaction is negative, while that for the main reaction is positive (deinsertion). The main reaction current, although several orders of magnitudes lower than that for the side reaction (in absolute value), is not zero because of the ohmic drop

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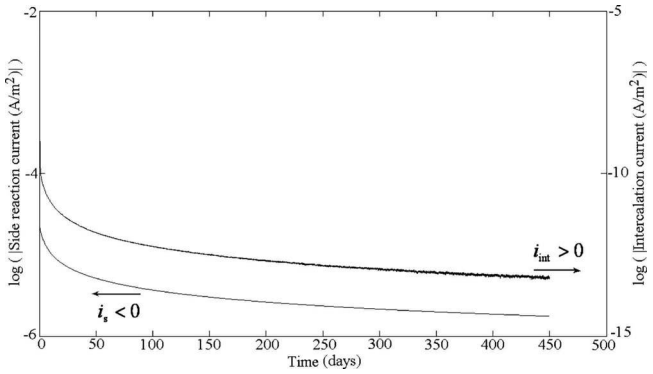

Figure 5. Simulation of side-reaction and intercalation current densities at the anode of an MP prototype cell under constant-voltage storage (3.9 V) at 30°C (diffusion-limited case).

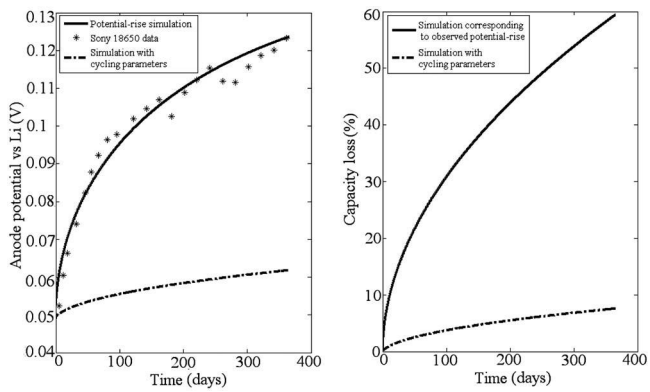

Figure 6. Anode potential rise and capacity loss simulations of a Sony 18650 cell during OCV storage at room temperature with an initial SOC = 99% (data taken from Ref. 6).

within the SEI, which is mainly caused by the side-reaction current ($R_{SEI}|i_t| \approx R_{SEI}|i_s|$) and which yields a positive overpotential for the main reaction.

OCV storage.— OCV storage of Li-ion batteries leads to a cell-voltage decrease together with a capacity loss generally composed of a reversible and an irreversible part. The causes for the reversible self-discharge are not generally clear. For instance, Yazami and Reynier suggested the formation of a metastable electron-ion-solvent complex that forms and adsorbs on the graphite during OCV storage.[34] Upon cell discharge, the complex is stable enough and accounts for the capacity loss. Upon cell recharge (graphite lithiation), the complex dissociates, and the capacity is thus recovered. According to Spotnitz,[35] acidic impurities could dissolve some of the anode SEI species that could be further reoxidized at the positive electrode, thus accounting for a reversible capacity loss as well. The reversible part of the capacity fade during OCV storage is not considered in the present model; however, using the experimental data provided by Ramasamy et al.,[6] a parameter-sensitivity analysis of capacity loss during storage is carried out. Ramasamy et al. provided the experimental data for the potential rise of a graphite electrode of a Sony 18650 cell upon OCV storage during a year at 25°C.[6] As for the other modes investigated, the capacity loss was assumed to be solely caused by the solvent decomposition leading to the growth of a SEI and therefore was totally irreversible. Because we have shown that the model parameters for a diffusion-limited regime of the side reaction are valid for aging upon cycling and upon storage at constant potential, the same set was also used for the simulation of OCV storage data. Besides, a new set of model parameters, given in Table II, was obtained from a refinement of the OCV storage data. The results of the two types of simulation are displayed in Fig. 6. No experimental data are reported in Ref. 6 for the capacity loss, which is why only the simulation is presented. Although the new set of refined parameters (Table II) is able to predict the potential-rise data, the corresponding simulated capacity fade amounts to 60%, which seems high. For comparison, Spotnitz reported an irreversible capacity loss of 17% for the same type of cell after 1 year of storage at 20°C.[35] The diffusion coefficient (3.7×10^{-19} m^2/s) is nearly 2 orders of magnitude larger than that obtained for the cycling and constant-potential storage situations. When the parameters obtained from cycling and constant-potential aging are used for the OCV storage, the potential rise is largely underestimated, while the capacity loss is in a more reasonable range (8%). From these trends, it is proposed that the cycling and constant-potential parameters are representative of the irreversible capacity loss during OCV storage. Indeed, if a mechanism such as the one proposed by Spotnitz is assumed,[35] reversible capacity fade cannot be observed during cycling and constant-potential storage, and therefore only an irreversible capacity fade is expected.

SEI quality and capacity fade.— The SEI porosity is believed to have quite a strong influence on Li-ion batteries capacity fade for both aging cycling and storage. It is a key factor in the quality of the SEI and consequently controls the rate of capacity fade of a battery. It is well-known that the SEI film has a limited elasticity and is prone to crack. According to Aurbach et al.,[7] Li-ion insertion into graphite particles during charge causes an increase in the lattice volume that may partially break the SEI film and therefore alters its passivity by exposing more of the underlying carbon to the electrolyte.

A fracture model for the SEI evolution that could be used in a physics-based capacity-fade model has not been proposed so far. Nevertheless, the porosity of the SEI was involved in the model developed in this study, and even though porosity changes during aging were not accounted for, it is worth looking at the effects of SEI-film porosity on the SEI growth rate. Using the parameters refined for the capacity loss during cycling and constant-potential storage (diffusion-limited case, Table II), a sensitivity analysis of the porosity effect on the capacity fade was performed by simulating case studies for which the SEI is either of high (1% porosity) or low (10% porosity) quality. Quality-signature curves that relate the storage time, the SOC, and the graphite potential to each other are presented in Fig. 7. The anode with a high-quality SEI (1% porosity) loses only 9% of initial capacity (SOC ≈ 90%) after 10 years of storage, while the low-quality SEI suffers from much more lithium exhaustion after the same amount of time (SOC ≈ 68%). After

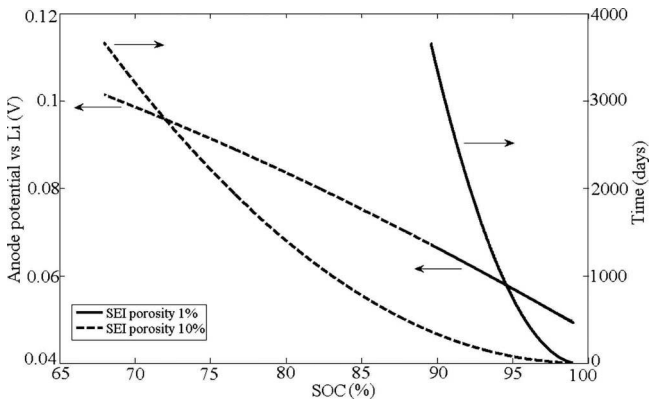

Figure 7. Quality-signature plots for an SEI with 1 or 10% porosity, simulated from graphite OCV storage at room temperature with an initial SOC of 99% using the aging parameters from the diffusion-limited case.

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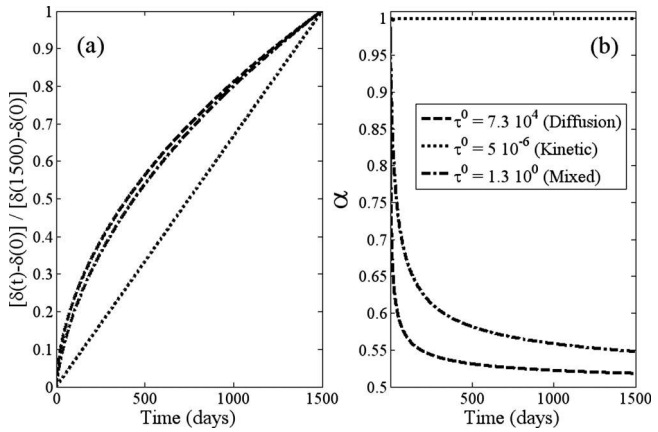

Figure 8. Dimensionless SEI thickness increase (a) and evolution of the growth law exponent (b) during the constant-potential storage of graphite electrode (0.064 V vs Li electrode) at room temperature for three different regimes of growth [diffusion-limited ($D_{EC} = 6.8 \times 10^{-21}$ m^2/s, $k_{f,s} = 10^{-7}$ m/s), kinetic-limited ($D_{EC} = 1.3 \times 10^{-15}$ m^2/s, $k_{f,s} = 1.36 \times 10^{-12}$ m/s), and mixed regime ($D_{EC} = 13.6 \times 10^{-21}$ m^2/s, $k_{f,s} = 3.4 \times 10^{-12}$ m/s)].

10 years, the potential reaches 0.07 V vs Li electrode in the case of the high-quality SEI and 0.1 V vs Li electrode for the low-quality one.

SEI growth and evolution of the growth law exponent.— In order to look at the SEI growth law in more detail, a dimensionless parameter $\tau = \delta k_{f,s}/D_{EC}$ is defined. Large values of τ are representative of diffusion-limited cases, while low and intermediate values of τ are for kinetic-limited and mixed regimes, respectively. Note that although changes in the growth regime may occur at long aging times due an increase of δ, here we focus on a comparative study of the relative values of $k_{f,s}$ and D_{EC}, and therefore $\tau^0 = \delta^0 k_{f,s}/D_{EC}$ is introduced instead of τ. Aging simulations for 1500 days under either constant anode potential (0.064 V vs Li electrode) or OCV ($SOC^0 = 0.99$) were run for the three SEI growth regimes ($\tau^0 = 7.3 \times 10^4$, $\tau^0 = 5 \times 10^{-6}$, and $\tau^0 = 1.3 \times 10^0$). Dimensionless SEI thicknesses $[\delta(t) - \delta(0)/\delta(1500) - \delta(0)]$ vs storage time are presented in Fig. 8a and 9a for constant-potential and OCV modes, respectively, while Fig. 8b and 9b exhibit the order of t in the SEI-growth equation $[\delta(t) - \delta(0) = bt^{\alpha}]$ as a function of time. As expected, no difference is observed between the two modes of operation for the diffusion-limited case; the SEI grows following a parabolic growth law, with the order α that decreases quickly at short times, probably because a solvent concentration needs to build up before solvent-diffusion limitations occur, and which eventually levels off toward a value close to 0.5. However, the kinetic-limited case is greatly affected by the operation mode. SEI growth is linear with time during potentiostatic aging, whereas it bends over with time during OCV mode, which stems from the anode potential change with time that leads to a decrease of the overpotential for the side reaction. The mixed regime lies between the two extreme cases, although it looks closer to the diffusion-limited one. Note that a similar parabolic growth behavior was reported by Peled et al., which was due to electron transport limitation in SEIs forming on lithium metal electrodes.[14] Later on, a more detailed study of the growth law was reported by Christensen and Newman for SEIs having a nonzero electron mobility as well.[15]

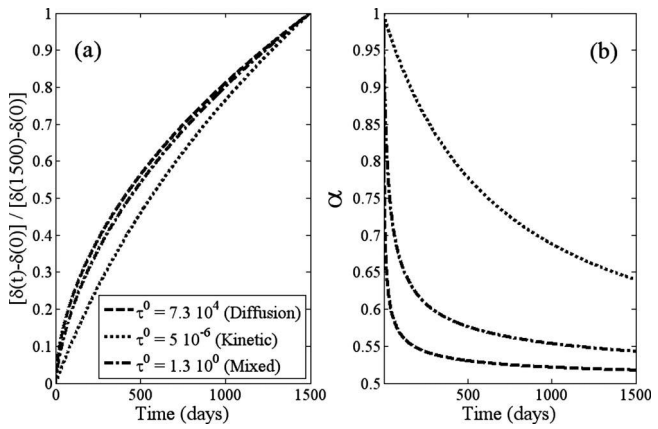

Figure 9. Dimensionless SEI thickness increase (a) and evolution of the growth law exponent (b) during the OCV storage of graphite electrode ($SOC^0 = 0.99$) at room temperature for three different regimes of growth [diffusion-limited ($D_{EC} = 6.8 \times 10^{-21}$ m^2/s, $k_{f,s} = 10^{-7}$ m/s), kinetic-limited ($D_{EC} = 1.3 \times 10^{-15}$ m^2/s, $k_{f,s} = 1.36 \times 10^{-12}$ m/s), and mixed regime ($D_{EC} = 13.6 \times 10^{-21}$ m^2/s, $k_{f,s} = 3.4 \times 10^{-12}$ m/s)].

Conclusion

A multimodal physics-based aging model that can be used both in cycling and storage condition was developed. The model is based on the continuous, small-scale growth of an SEI layer on the surface of the anode active particles. The solvent decomposition kinetics and solvent diffusion through the SEI layer were both considered in the model. The $LiCoO_2$/graphite cell was chosen as a case study, and using the various experimental data from the literature, the aging of the cell upon both cycling and storage was simulated.

In cycling mode, the contribution of the anode SEI growth to the total capacity fade in the cell was first estimated together with the interfacial resistance increase. From simulation, it was shown that both diffusion- and kinetic-limited regimes were able to meet the estimated values. More experimental data are required to assess whether this is more kinetic- or diffusion-limited or to refine simultaneously the two side-reaction parameters (i.e., mixed regime). The same set of side-reaction parameters was used to simulate the capacity fade during constant-potential storage, and contrary to the cycling mode, the kinetic-limited case failed to describe the capacity fade. This proves that SEI growth is more likely to be a diffusion-limited process rather than a kinetic-limited one. For OCV storage, it was shown that the set of model parameters refined for the other modes could not satisfy the experimental data, pointing out that part of the capacity loss is reversible.

The SEI porosity was shown to be a critical parameter that reflects the quality of the SEI in terms of passivation properties. Using the simulation results, it was shown that an SEI layer with 10% porosity leads to a capacity fade more than 3 times higher than for an SEI with 1% porosity.

Finally, the growth law of SEI was studied for the two storage modes in a more systematic way by looking at different extreme cases. While the SEI grows according to a parabolic law regardless of the storage mode when the growth is under solvent-diffusion limitations, the kinetic-limited growth of the SEI leads to different behaviors depending on the operation mode.

Besides the work on anodes, the necessity of modeling structural degradations in the cathode is an important step in the development of battery-life prediction models. Addition of thermal concerns to the present model and considering composition and porosity changes of the SEI as well as SEI dissolution could be the subject of a future work.

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Appendix

For the graphite electrode[27]

$$U_- = 0.7222 + 0.1387x + 0.029x^{0.5} - \frac{0.0172}{x} + \frac{0.0019}{x^{1.5}} + 0.2808\exp(0.9 - 15x) - 0.7984\exp(0.4465x - 0.4108)$$

For the $LiCoO_2$ electrode[27]

$$U_+ = \frac{-4.656 + 88.669y^2 - 401.119y^4 + 342.909y^6 - 462.471y^8 + 433.434y^{10}}{-1 + 18.933y^2 - 79.532y^4 + 37.311y^6 - 73.083y^8 + 95.96y^{10}}$$

List of Symbols

A	total interfacial surface area of the electrode, m^2
b	constant in the SEI growth law, $\delta(t) - \delta(0) = bt^{\alpha}$
c_{EC}	solvent concentration in the SEI film, mol/m^3
$c_{EC,sln}$	solvent concentration in the bulk of the electrolyte, mol/m^3
c_{Li}	concentration of Li inside the active material, mol/m^3
c_{Li}^{max}	maximum concentration of Li inside the active material, mol/m^3
D_{EC}	solvent diffusion coefficient inside the SEI layer, m^2/s
D_{Li}	lithium diffusion coefficient inside active particles, m^2/s
F	Faraday's constant, 96,487 C/mol
i_{int}	intercalation current density referred to interfacial surface area of anode, A/m^2
i_s	side-reaction current density referred to interfacial surface area of anode, A/m^2
i_t	total current density referred to interfacial surface area of anode, A/m^2
K_{EC}	partition coefficient of solvent between the electrolyte and the SEI
$k_{f,s}$	rate constant of side reaction, m/s
$k_{int}^{0,app}$	apparent rate constant for lithium intercalation reaction, m/s
M_{SEI}	molecular weight of SEI, kg/mol
N_{EC}	solvent flux density, $mol/(m^2\ s)$
r	radial distance in active particle and across the SEI, m
R	active particles radius (m) and ideal gas constant, 8.314 J/(mol K)
R_{SEI}	SEI film resistance referred to interfacial surface area of anode, $\Omega\ m^2$
t	time, s
T	temperature, K
U	equilibrium potential vs Li electrode, V
v	SEI film velocity, m/s
x	lithium stoichiometry in anode
y	lithium stoichiometry in cathode
α	power law exponent for SEI growth
β	cathodic charge-transfer coefficient for the intercalation reaction
β_s	cathodic charge-transfer coefficient for the side reaction
δ	SEI film thickness, m
ε_{SEI}	SEI film porosity
Φ_1	potential of the electron-conductive phase, V
Φ_2	potential of the ion-conductive phase, V
κ_{SEI}	SEI ionic conductivity, S/m
ρ_{SEI}	SEI density, kg/m^3
τ	dimensionless parameter determining SEI growth regime

Subscript

s	side reaction
+	variables referred to cathode
−	variables referred to anode

Superscript

0	initial value

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